

Visual Simulation of Tire Smoke

Tomoya Tamagawa
Toyo University
Tokyo, Japan
3F102200077@iniad.org

Tomokazu Ishikawa
Toyo University
Prometech CG Research
Tokyo, Japan
Tomokazu.Ishikawa@gmail.com



Figure 1: Tire smoke during burnout.

ABSTRACT

In recent years, the realism of video expression by 3DCG has increased remarkably. In this study, we propose a CG representation method of tire smoke based on physical simulation. Although tire smoke has appeared in various video works and 3DCG games, the mechanism of how tire smoke is generated has not been fully elucidated. We calculate the generation and conduction of heat from the rotation speed and friction between the tire and the ground, and construct a model of tire smoke generation from the viewpoint of statistical mechanics. The parameters in the mathematical model are set from actual images and observations, and we confirm that the phenomenon of tire smoke can be reproduced.

ACM Reference Format:

Tomoya Tamagawa and Tomokazu Ishikawa. 2022. Visual Simulation of Tire Smoke. In *SIGGRAPH Asia 2022 Posters (SA '22 Posters)*, December 06-09, 2022. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3550082.3564167>

1 INTRODUCTION

In recent years, the use of 3DCG to express images has become significantly more realistic. Various technologies have been developed and used for video expression, ranging from representation of unrealistic phenomena to representation based on physical simulations [Fedkiw et al. 2001; Stam 1999]. In this study, we propose a method for representing tire smoke based on physical simulation. Transparent mapping has been the main method used in existing representations. Tire smoke has been represented in various video

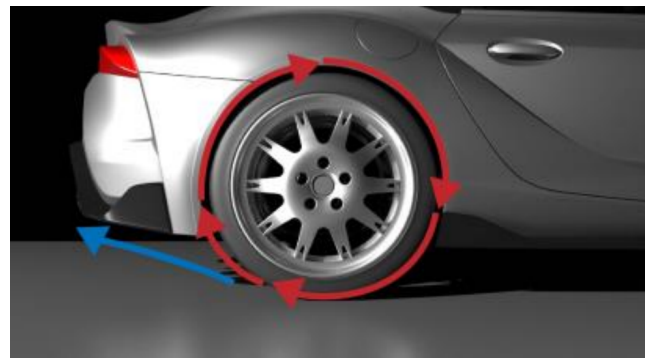


Figure 2: Airflow around tire house.

works and 3DCG games, but tire smoke in real world is a complex phenomenon. In most images, the smoke is emitted only from the ground surface of the tire, but in reality, tire smoke appears from the entire tire, and the smoke is caught by the air currents around the tire. Therefore, we considered reproducing the phenomenon of tire smoke more precisely in terms of tire temperature and rotational speed.

2 TYRE SMOKE

In this section, we discuss one of the key components of tire smoke: tire temperature variation. When a tire spins, the rubber of a tire heats up to over 400°C due to friction between the tire and the asphalt. Because of this, the rubber tread melts, vaporizing the chemicals and oils inside and releasing their molecules into the air. The vaporized molecules quickly cool and condense in the air, forming a visible white “smoke”.

There are two objects that interfere with the generation of tire smoke: the ground and the tire house. The tire house is the second

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

SA '22 Posters, December 06-09, 2022, Daegu, Republic of Korea

© 2022 Copyright held by the owner/author(s).

ACM ISBN 978-1-4503-9462-8/22/12.

<https://doi.org/10.1145/3550082.3564167>

most influential factor in reproducing tire smoke after tire temperature variation. Figure 2 shows a schematic of airflow around a tire house. The red arrows indicate airflow circulating in the tire house, and the blue arrows indicate airflow from the tire house toward the rear of the leak. In this study, the airflow in the tire house was reproduced by fluid simulation using the tire house as a fixed boundary condition.

3 OUR METHOD

The heat quantity Q_f due to dynamic friction force is considered into the same as the work done by the dynamic friction force and is calculated by $Q_f = \mu N \Delta x$. A real tire used as reference in modeling had a horizontal length of 128 mm and a tire width of 215 mm.

Based on Fourier's law, consider the following time evolution equation for the ground temperature T_g and the tire temperature T_t ,

$$T_t(t + \Delta t) = \begin{cases} T_t(t) - \frac{k_t}{C_t} (T_t(t) - T_{air}) \Delta x \Delta t & (1) \\ T_t(t) + \frac{Q_f}{C_t} - \frac{k_t}{C_t} (T_t(t) - T_g) \Delta x \Delta t, & (2) \end{cases}$$

$$T_g(t + \Delta t) = \begin{cases} T_g(t) - \frac{k_g}{C_g} (T_g(t) - T_{air}) \Delta x \Delta t & (3) \\ T_g(t) + \frac{Q_f}{C_g} - \frac{k_g}{C_g} (T_g(t) - T_t) \Delta x \Delta t. & (4) \end{cases}$$

C is heat capacity, k is thermal conductivity, and subscripts indicate material as same as temperature. T_{air} is the temperature of the air. x is the horizontal distance of the tire's ground contact area. The actual tire was measured to have an area of 27,520 mm² installed with the ground. The equation for the temperature change when the tire is spinning is branched into Eq. (1) and Eq. (3) when the tire is not on the ground and Eq. (2) and Eq. (4) touching on the ground.

The results of numerical simulations using from Eq. (1) to Eq. (4) are shown in Figure 3 for the case of burnout (the car doesn't move, only the tires rotate) and the case of normal driving. The vertical and horizontal axes are temperature (°C) and time (s), respectively. In the case of burnout, the tire temperature rises due to heat conduction with the asphalt and eventually converges. This result generally agrees with the measurement results taken by thermography in the real world [Explained 2017].

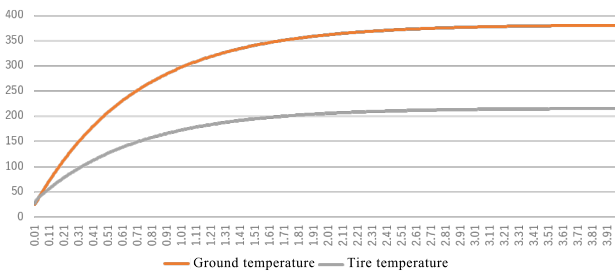


Figure 3: Numerical simulation of temperature change



Figure 4: Calculation result after 2.5 seconds from tire burnout.

We assume that the smoke generation follows a Maxwell-Boltzmann distribution.

$$\frac{dN}{N} = 4\pi v^2 \left(\frac{m}{2\pi k_B T} \right)^{\frac{3}{2}} e^{-\frac{mv^2}{2k_B T}} dv, \quad (5)$$

where v is the velocity of the molecule, m is the mass of the molecule, k_B is Boltzmann's constant, and T is the temperature. In case of tire smoke, we consider that something with a constant velocity (kinetic energy) is generated as smoke.

4 RESULTS AND CONCLUSION

Rendering was performed from a fluid simulation in Houdini using a PC with CPU: AMD Ryzen 7 2700 @3.2GHz and RAM: 48GB. With a rendering resolution of 1280 x 720, the computation time was 7 minutes per frame. Figure 4 shows the image 2.5 second after the tires were burnout. Tire smoke is beginning to appear. The temperatures of the tire and the ground at this time are 172.4°C and 296.4°C, respectively.

In this paper, the temperature change of the tire was simulated and visualized based on the tire heat conduction equation. The simulation showed that tire smoke can be reproduced as in the real world. Future work includes measuring actual tire temperatures and evaluating the error between the numerical simulation and the actual tire temperature. We plan to further reduce computational cost to achieve real-time rendering and make it usable in games.

ACKNOWLEDGMENTS

This work was supported by Toyo University Top Priority Research Program.

REFERENCES

- Engineering Explained. 2017. Huge Burnout With A Thermal Camera. <https://www.youtube.com/watch?v=MJ1NK02CL6g> (accessed 16 Aug. 2022).
- Ronald Fedkiw, Jos Stam, and Henrik Wann Jensen. 2001. Visual Simulation of Smoke. In *Proceedings of the 28th Annual Conference on Computer Graphics and Interactive Techniques (SIGGRAPH '01)*. Association for Computing Machinery, New York, NY, USA, 15–22. <https://doi.org/10.1145/383259.383260>
- Jos Stam. 1999. Stable Fluids. In *Proceedings of the 26th Annual Conference on Computer Graphics and Interactive Techniques (SIGGRAPH '99)*. ACM Press/Addison-Wesley Publishing Co., USA, 121–128. <https://doi.org/10.1145/311535.311548>